

- 4 (a) Starting from the definitions of $\cosh$ and $\sinh$ in terms of exponentials, prove that

$$\cosh^2 x - \sinh^2 x = 1. \quad [3]$$

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- (b) Show that $\frac{d}{dx}(\tan^{-1}(\sinh x)) = \operatorname{sech} x$. [3]

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- (c) Sketch the graph of $y = \operatorname{sech} x$, stating the equation of the asymptote. [2]

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- (d) By considering a suitable set of n rectangles of unit width, use your sketch to show that

$$\sum_{r=1}^n \operatorname{sech} r < \tan^{-1}(\sinh n). \quad [3]$$

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- (e) Hence state an upper bound, in terms of π , for $\sum_{r=1}^{\infty} \operatorname{sech} r$. [1]

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